TEACHER COPY · EXAMPLES AND ANSWERS

Bridge to Year 9

Year 8 mixed-ability class → Year 9 Standard (Stage 5 core) · 7 × 50-minute lessons · Mandelbrot series

Overview

BOOKLET PAGES	SECTION	CORE OUTCOMES	GOLD ROUND
3–4	Pre-test (20 marks, 20 min)	One question per lesson; use it to decide where each student starts.	
5–7	Lesson 1: Number skills	MA4-INT-C-01 · MA4-FRC-C-01	Gold: harder Stage 4 problems (no Stage 5 content)
8–10	Lesson 2: Algebra basics	MA4-ALG-C-01	Gold: harder Stage 4 problems (no Stage 5 content)
11–13	Lesson 3: Expanding & factorising	MA4-ALG-C-01	Gold: harder Stage 4 problems (no Stage 5 content)
14–16	Lesson 4: Indices & square roots	MA4-IND-C-01	Gold: harder Stage 4 problems (no Stage 5 content)
17–19	Lesson 5: Pythagoras' theorem	MA4-PYT-C-01	Gold: harder Stage 4 problems (no Stage 5 content)
20–22	Lesson 6: Equations & formulas	MA4-EQU-C-01	Gold: harder Stage 4 problems (no Stage 5 content)
23–25	Lesson 7: Area & volume	MA4-ARE-C-01 · MA4-VOL-C-01	Gold: harder Stage 4 problems (no Stage 5 content)
26–27	Post-test (20 marks, 20 min)	Parallel form of the pre-test.	

Notes for the teacher

- **Levels.** Bronze = support (nearly everyone unaided; some questions show the first step), Silver = core Stage 4, Gold = harder multi-step Stage 4 problems. No Stage 5 content is included. MAO-WM-01 applies throughout.
- **Examples are teacher-led.** The student copy shows each example question with blank space so you can work it live; the full working is in the box for each lesson below.
- **Stage 4 only.** Expanding and factorising use single brackets with a positive term outside; negative coefficients, binomial products, surds and changing the subject (all Stage 5) are left for Year 9.
- **Interleaving.** 35 of the 195 drill questions (18%; 13%–24% per lesson) come from an earlier lesson (chip ↻ L1–L6) or from Year 8 topics not taught in the booklet (integers, fractions, percentages, ratios, linear relationships, equations).
- **Timing.** 254 questions in total (34–38 per lesson). In 50 minutes most students will finish Bronze and most of Silver; Gold and any leftovers suit early finishers or homework.
- **Calculator allowed.** Answers that need rounding say how many decimal places; answers were rounded only at the last step.
- **Every answer below was checked by computer algebra** (expansions, factorisations, solutions, substitutions and all numerical values).

1 Number skills

EXAMPLES (WORKED)

Example 1 Evaluate $-6 + 4 \times (-3)$.

$$= -6 + (-12)$$
$$= -18$$

Example 2 Evaluate $\frac{2}{3} + \frac{3}{4}$.

$$= \frac{8}{12} + \frac{9}{12}$$
$$= \frac{17}{12} = 1\frac{5}{12}$$

Example 3 Increase $\$240$ by 15%.

$$= 240 \times 1.15$$
$$= \$276$$

Bronze round

1 3
2 -9
3 6
4 -20
5 6
6 11
7 14
8 $\frac{3}{4}$
9 $\frac{7}{12}$
10 $\frac{2}{5}$
11 9
12 20
13 3.5
14 60%
15 $2 : 3$ ↻ Ratios
16 50° ↻ Angles

S Silver round

17 -24 **18** 32 **19** $\frac{5}{12}$ **20** $4\frac{1}{6}$

21 $\frac{3}{2} = 1\frac{1}{2}$ **22** 3 **23** $\$92$ **24** $\$84$

25 25% **26** 0.72 **27** 18 **28** 12.68

29 $\$20$ and $\$40$ ↻ Ratios **30** 6 ↻ Data

31 $95 \times 0.8 = \$76$

32 $-4 + 11 - 3 = 4\,^\circ\text{C}$

G Gold round

33 No: $100 \times 1.1 = 110$, then $110 \times 0.9 = \$99$

34 $\frac{5}{12}$ ($\frac{4}{12}$ and $\frac{6}{12}$)

35 $68 \div 0.85 = \$80$

36 $-3 \times 9 \div 9 = -3$

37 3 and -5

38 $\$2.50$ per 100 g vs $\$2.32$ per 100 g: the 750 g pack

2 Algebra basics

EXAMPLES (WORKED)

Example 1 Simplify $5x + 3y - 2x + 7y$.

$$= 5x - 2x + 3y + 7y$$
$$= 3x + 10y$$

Example 2 Simplify $4a \times 3ab$.

$$= 4 \times 3 \times a \times a \times b$$
$$= 12a^2b$$

Example 3 Evaluate $2a^2 - 3b$ when $a = -3$ and $b = 4$.

$$= 2(-3)^2 - 3(4)$$
$$= 18 - 12 = 6$$

B Bronze round

1 $10a$ **2** $5m$ **3** $8x + 2y$ **4** $12k$

5 $10ab$ **6** $4y$ **7** $4a$ **8** $5p + 2q$

9 $3m^2$ **10** $5a + 5b$ **11** 17 **12** $n + 5$

13 $3y$ **14** 5 ↻ L1 **15** 27 ↻ L1 **16** 2

S Silver round

17 $2x^2 + 7x$ **18** $ab + 2a$ **19** $6x^2y$ **20** $3a$

21 $\frac{2x}{3}$ **22** $12mn$

23 -1 **24** 4 **25** 2 **26** -30

27 $\$69$ ↻ L1 **28** -11 ↻ Linear

29 a) $14n + 6$ b) $\$48$

30 e.g. $x = 2$, $y = 1$: $3x + 2y = 8$ but $5xy = 10$. They are unlike terms.

G Gold round

31 a) 152 b) No: $3n + 2 = 100$ gives $n = 32\frac{2}{3}$, not a whole number

32 Perimeter $6a$ cm; area $2a^2$ cm^2

33 $56.25 - 37.5 + 6.25 = 25$

34 $4 + 2.5k$; $\$34$

35 any number between 0 and 1, e.g. $x = 0.5$: $0.25 < 0.5$

36 4

3 Expanding & factorising

EXAMPLES (WORKED)

Example 1 Expand $4(2a + 3)$.

$$= 4 \times 2a + 4 \times 3$$
$$= 8a + 12$$

Example 2 Expand and simplify $3(x + 2) + 2(x + 5)$.

$$= 3x + 6 + 2x + 10$$
$$= 5x + 16$$

Example 3 Factorise $12x + 18$.

$$\text{HCF} = 6$$
$$= 6(2x + 3)$$

B Bronze round

1 $3x + 12$ **2** $5a - 10$ **3** $6m + 2$ **4** $4x + 20$
5 $x^2 + 6x$ **6** $ab - 3a$ **7** 5 L1 **8** $12a^2$ L2
9 6 **10** $3(x + 4)$ **11** $5(a - 4)$ **12** $7(y - 2)$
13 $3(2x + 3)$ **14** $x(x + 3)$

S Silver round

15 $6a^2 + 10ab$ **16** $10m - 5m^2$ **17** $7x + 10$ **18** $x^2 + 5x - 8$
19 $11y + 8$ **20** $\frac{5}{6}$ L1
21 $2(2m + 3)$ **22** $x(x - 7)$ **23** $2a(2a + 5)$ **24** $3m(3n - 2)$
25 $5pq(3p + 4q)$ **26** $3(x^2 + 2x + 3)$ **27** \$17 L1 **28** 14 L2

29 He did not multiply the 5 by 2: $2x + 10$

30 $(2x + 5)$ cm

31 $17(23 + 77) = 17 \times 100 = 1700$

G Gold round

32 $9x + 9 = 9(x + 1)$

33 $6ab(2a + 3b - 1)$

34 e.g. $4(3x + 2)$, $2(6x + 4)$, $2(3x + 4) + 6x$ (any correct)

35 $n(n + 1)$: two consecutive whole numbers, so one of them is even

36 $2(x + 4) = 2x + 8$ m^2

37 No: $4x - 6$ still has a common factor of 2; $4x(2x - 3)$

4 Indices & square roots

EXAMPLES (WORKED)

Example 1 Evaluate 3^4.

$$= 3 \times 3 \times 3 \times 3$$
$$= 81$$

Example 2 Write $2^3 \times 2^4$ as a single power.

$$= 2^{3+4}$$
$$= 2^7$$

Example 3 Write 72 as a product of prime factors.

$$72 = 2 \times 2 \times 2 \times 3 \times 3$$
$$= 2^3 \times 3^2$$

B Bronze round

1 5^3 **2** 32 **3** 1000 **4** 25
5 1 **6** 9 **7** 4 **8** 2^7
9 5^6 **10** 3^8 **11** $20xy$ L2 **12** $4(2x - 5)$ L3
13 $2^2 \times 3^2$ **14** $2^3 \times 3$ **15** $\frac{7}{10}$ L1

S Silver round

16 49 **17** 4 **18** 9 **19** 21
20 24 **21** 72 **22** $n = 7$ **23** $n = 7$
24 $2 \times 3^2 \times 5$ **25** $2^2 \times 3^3 \times 5$ **26** 7 and 8 **27** 7.1
28 \$200 L1 **29** $6x + 15$ L3

30 False: $2^3 \times 2^4 = 2^7 = 128$, but $4^7 = 16\,384$. Keep the base; add the indices.

31 $2^{10} = 1024$

32 $5^3 \div 5^3 = 5^0$ by the index law, and any number divided by itself is 1.

G Gold round

33 $2^{30} = 8^{10}$ and $3^{20} = 9^{10}$, so 3^{20} is larger

34 $2^{2n} = 2^{10}$, so $n = 5$

35 $2 \times 3 \times 7 = 42$

36 2: $72 = 2^3 \times 3^2$, so $72 \times 2 = 144 = 12^2$

37 2

38 $\sqrt{200} \approx 14.1$ cm

5 Pythagoras' theorem

EXAMPLES (WORKED)

Example 1 Find c if the shorter sides are 5 cm and 12 cm.

$$c^2 = 5^2 + 12^2 = 169$$
$$c = \sqrt{169} = 13 \text{ cm}$$

Example 2 Find x if the hypotenuse is 9 cm and one side is 5 cm (1 d.p.).

$$x^2 = 9^2 - 5^2 = 56$$
$$x = \sqrt{56} \approx 7.5 \text{ cm}$$

Example 3 Is a triangle with sides $7, 24, 25$ right-angled?

$$7^2 + 24^2 = 625$$
$$25^2 = 625 \checkmark \text{ Yes}$$

B Bronze round

1 10 cm **2** 15 cm **3** 5 cm **4** $\sqrt{74} \approx 8.6$ cm

5 PR **6** 7 L4 **7** 8.60 L1 **8** 100

9 13 cm **10** $\sqrt{13} \approx 3.6$ cm **11** 8 cm **12** 25

S Silver round

13 12 cm **14** 24 cm **15** $\sqrt{84} \approx 9.2$ cm **16** 10 cm

17 Yes: $81 + 1600 = 1681 = 41^2$ **18** No: $36 + 81 = 117 \neq 121$ **19** Yes: $400 + 441 = 841$ **20** $x(x + 4)$ L3

21 17 L4 **22** 25 L2 **23** 12 cm **24** 50 cm

25 \$36 L1

26 $\sqrt{25 - 1.96} = 4.8$ m

27 $\sqrt{8425} \approx 91.8$ cm

28 650 m

G Gold round

29 $\sqrt{9 - 7.84} = \sqrt{1.16} \approx 1.1$ m

30 hypotenuse 10; perimeter 24 cm

31 height $\sqrt{169 - 25} = 12$ cm; area 60 cm^2

32 diagonal 130 m; $170 - 130 = 40$ m shorter

33 e.g. $6, 8, 10$: $36 + 64 = 100$; $9, 12, 15$: $81 + 144 = 225$

34 $\sqrt{2500 - 900} = 40$ m

6 Equations & formulas

EXAMPLES (WORKED)

Example 1 Solve $5x - 7 = 18$.

$$5x = 25$$
$$x = 5$$

Example 2 Solve $7x - 3 = 4x + 9$.

$$3x - 3 = 9$$
$$3x = 12, \quad x = 4$$

Example 3 Solve $2x^2 = 50$.

$$x^2 = 25$$
$$x = \pm 5$$

B Bronze round

1 $x = 6$ **2** $y = 1$ **3** $m = 7$ **4** $k = 15$

5 $x = 4$ **6** $a = 7$ **7** $x = 6$ **8** $n = 12$

9 $x = \pm 6$ **10** $x = 6$ **11** $6x - 12$ L3 **12** 243 L4

13 5 L5 **14** $x = 4$

S Silver round

15 $x = 4$ **16** $a = -4$ **17** $x = 12$ **18** $x = 3$

19 $x = 6$ **20** $x = 19$ **21** $x = \pm 4$ **22** $x = \pm\sqrt{8} \approx \pm 2.83$

23 \$13.50 L1 **24** $\frac{7}{12}$ L1

25 $t = 5$ **26** $h = 7.5$ **27** $r \approx 3.99$ **28** $C = 35$

29 $4n - 7 = 2n + 9$, so $n = 8$

30 $4.50 + 2.20d = 37.50$, $d = 15$ km

G Gold round

31 $80 + 65h = 307.50$, $h = 3.5$ hours

32 $4w + 6 = 42$, $w = 9$; 9 cm by 12 cm

33 $x^2 = 81$, $x = \pm 9$

34 $3n + 3 = 138$; $45, 46, 47$

35 $r = \sqrt{200 \div \pi} \approx 8.0$ cm

36 $20 + 0.5g = 12 + g$, $g = 16$ GB

7 Area & volume

EXAMPLES (WORKED)

Example 1 Trapezium: $a = 6$, $b = 10$, $h = 5$ (cm). Find the area.

$$A = \tfrac{5}{2}(6 + 10) = 40 \text{ cm}^2$$

Example 2 Circle: radius 3 cm. Find the area (1 d.p.).

$$A = \pi \times 3^2 \approx 28.3 \text{ cm}^2$$

Example 3 Cylinder: $r = 3$ cm, $h = 10$ cm. Find the volume (1 d.p.).

$$V = \pi \times 3^2 \times 10 \approx 282.7 \text{ cm}^3$$

B Bronze round

1 36 m² **2** 30 cm² **3** 60 cm² **4** 30 cm²

5 30 cm² **6** 78.5 cm² **7** 120 cm³ **8** 60 cm³

9 216 cm³ **10** $x = 5$ L6 **11** 10 L5 **12** 64 L4

13 49 cm² **14** 50.3 cm² **15** 31.4 cm³

S Silver round

16 $80 - 18 = 62$ cm² **17** $60 + 24 = 84$ cm² **18** $A = 21$; $V = 420$ cm³

19 39.3 cm² **20** 628.3 cm³ **21** 942.5 cm³ **22** 8 cm L6

23 375 m³ = 375 000 L **24** $4x + 12$ L3 **25** 42 m² **26** 120 cm³

27 7 cm L4

28 $A = 9$ m²; $V = 2.7$ m³

29 Two identical triangles fit together to make a rectangle (or parallelogram) b by h, so the triangle is $\frac{1}{2}bh$.

G Gold round

30 $\pi(25 - 16) \times 100 \approx 2827.4$ cm³

31 $h = 1000 \div (25\pi) \approx 12.7$ cm

32 $1000 - 40\pi \approx 874.3$ cm³

33 $\pi \times 1.44 \times 2 \approx 9.048$ m³ ≈ 9048 L

34 $\frac{5}{2}(2x + 4) = 45$, $x = 7$

35 It is multiplied by 4: $\pi(2r)^2 = 4\pi r^2$

Pre-test and post-test – marking guide (20 marks each)

Q	PRE-TEST ANSWER	POST-TEST ANSWER	MARK
1a	-14	-18	1
1b	$\frac{11}{12}$	$\frac{11}{10} = 1\frac{1}{10}$	1

1c	$\$92$	$\$72$	1
2a	$3x + 3y$	$4m + 2n$	1
2b	$12ab$	$10xy$	1
2c	-2	-11	1
3a	$3x + 12$	$5y - 10$	1
3b	$6a + 6$	$5m + 6$	1
3c	$6(2m + 3)$	$5(3k + 4)$	1
4a	5^3	3^3	1
4b	72	80	1
4c	$2^2 \times 3^2 \times 5$	$2^2 \times 3^2 \times 7$	1
5a	15 cm	25 cm	1
5b	8 cm	12 cm	1
5c	$\sqrt{65} \approx 8.1$ cm	$\sqrt{106} \approx 10.3$ cm	1
6a	$x = 5$	$x = 7$	1
6b	$a = 2$	$a = 4$	1
6c	$x = \pm 6$	$x = \pm 5$	1
7a	28 cm^2	40 cm^2	1
7b	141.4 cm^3	88.0 cm^3	1

Award the mark for the correct final answer; accept equivalent forms and correct rounding of a correct method. In Q3c a correct but not fully factorised answer scores 0, because factorising fully is the skill tested.